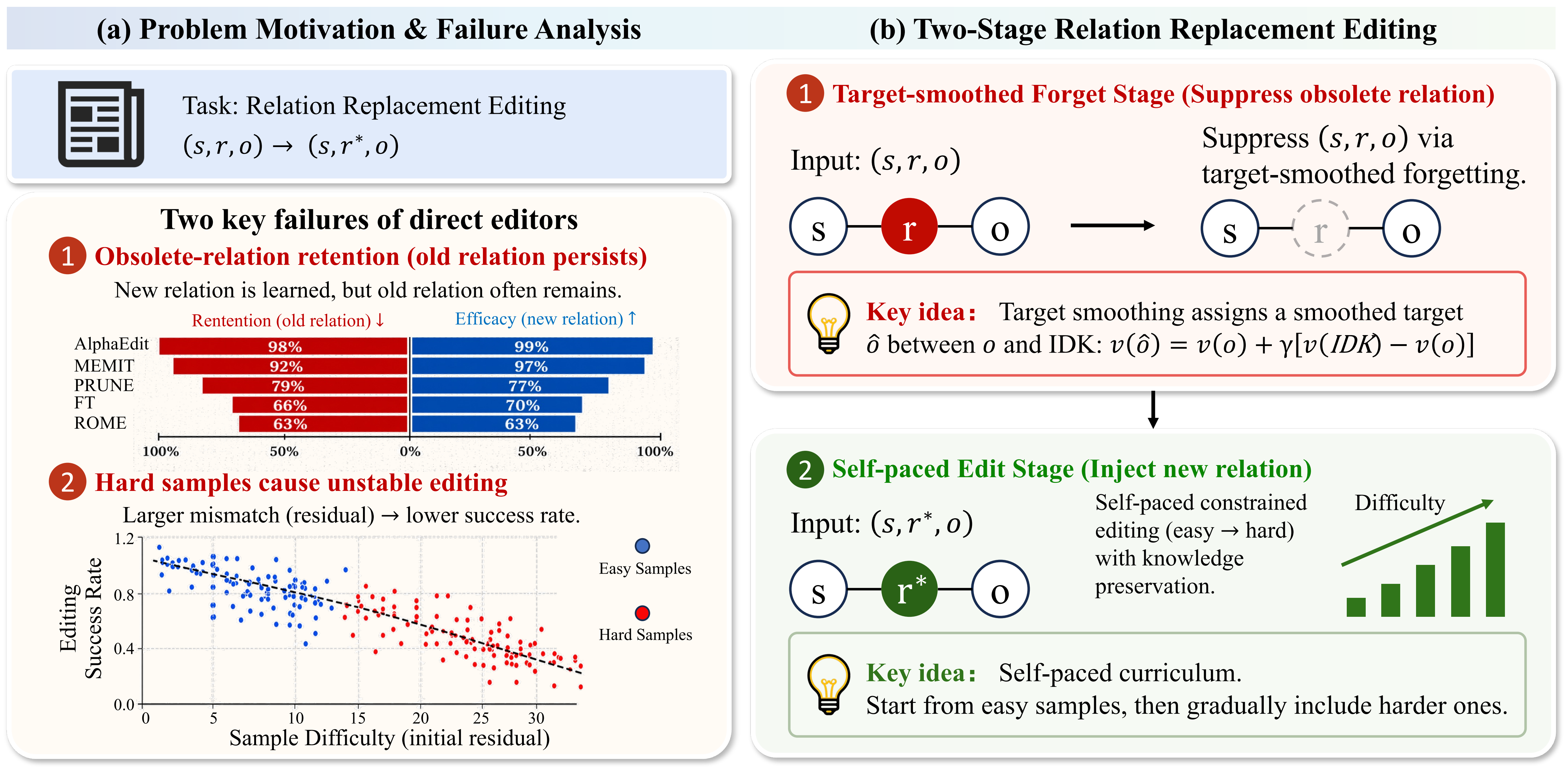

(a) Problem Motivation & Failure Analysis
Task: Relation Replacement Editing
$(s, r, o) \rightarrow (s, r^*, o)$
Two key failures of direct editors
1 Obsolete-relation retention (old relation persists)
New relation is learned, but old relation often remains.
Rentention (old relation) ↓
Efficacy (new relation) ↑
AlphaEdit 98% 99%
MEMIT 92% 97%
PRUNE 79% 77%
FT 66% 70%
ROME 63% 63%
100% 50% 0% 50% 100%
2 Hard samples cause unstable editing
Larger mismatch (residual) → lower success rate.
Editing Success Rate
1.2 0.8 0.4 0.0
0 5 10 15 20 25 30
Sample Difficulty (initial residual)
Easy Samples
Hard Samples
(b) Two-Stage Relation Replacement Editing
1 Target-smoothed Forget Stage (Suppress obsolete relation)
Input: (s, r, o)
Suppress (s, r, o) via target-smoothed forgetting.
s r o
s r o
Key idea: Target smoothing assigns a smoothed target $\hat{o}$ between o and IDK: $v(\hat{o}) = v(o) + \gamma[v(IDK) - v(o)]$
2 Self-paced Edit Stage (Inject new relation)
Input: (s, r^*, o)
s r^* o
Self-paced constrained editing (easy → hard) with knowledge preservation.
Difficulty
Key idea: Self-paced curriculum.
Start from easy samples, then gradually include harder ones.